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COMMON QUARTERLY EXAMINATION - 2025

STANDARD - XII

Reg.No. | | | | | | |

Time: 3.00 hrs

BIOLOGY

Marks:70

PART - I BIO - BOTANY

Marks:35

Section - I

I Answer all the questions: 8x1=8

1. Coleorhiza is found in
   a) Paddy b) Bean c) Pea d) Tridax
2. The genes controlling the seven pea characters studied by Mendel are known to be located on how many different chromosomes
   a) Seven b) Six c) Five d) Four
3. Assertion: (A) Gamma rays are generally use to induce mutation in wheat varieties
   Reason (R) Because they carry lower energy to non-ionize electrons from atom
   a) A and R are correct
   b) A is correct R is wrong
   c) A is wrong R is correct
   d) A and R are wrong
4. EcoRI cleaves DNA at
   a) AGGGTT b) GTATATC c) GAATTC d) TATAGC
5. Match the following:

Column A	Column B
1. Totipotency	(A) Reversion of mature cells into meristerm
2. Dedifferentiation	(B) Biochemical and structural changes of cells
3. Explant	(C) Properties of living cells develops into entire plant
4. Differentiation	(D) Selected plant tissue transferred to culture medium

6. Virus free plants are developed from
   a) Organ culture
   b) Meristem culture
   c) protoplast culture
   d) Cell suspension culture
7. Separation of proteins by the blotting technique
   a) Southern blotting
   b) Northern blotting
   c) Western blotting
   d) Agrose Gel Electroporosis
8. Which is popularly called "Terror of Bengal"
   a) Eichhornia crassipus
   b) Argemone mexicana
   c) Phyllium frondosum
   d) Carrausium morossus

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Section - II

II Answer any four of the following questions. 4x2=8

9. What is meant by Myrmecophilly.
10. What are Embriyoids
11. Define totipotency.
12. What is meant by Synonymous or silent mutation?
13. What is meant by Variation?
14. Write the advantages of Bt cotton.

Section - III

III Answer any three of the following questions. (Q.No.19 is compulsory) 3x3=9

15. Give an account on cryopreservation.
16. What is gene mapping? Write its uses.
17. Distinguish mound layering and air layering.
18. What do you know about the word pBR322?
19. What is Atavism, explain it with example?

Section - IV

IV Answer all the questions. 2x5=10

20. a) Bring out the inheritance of chloroplast gene with an Example.

(OR)

b) What is endosperm. Explain the types.

21. a) Write about the applications of plant tissue culture.

(OR)

b) Compare the various types of Blotting techniques.

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PART - II BIO - ZOOLOGY Marks:35

Section - I

I Answer all the questions: 8x1=8

1. The mode of sexual reproduction in bacteria is by
   a) Formation of gametes b) Endosperm formation
   c) Conjugation d) Loospore formation
2. Mammalian egg is
   a) Mesolecithal and non-cleidoic b) Microlecithal and non-cleidoic
   c) A lecithal and non-cleidoic d) A lecithal and cleidoic
3. Which one of the following groups includes sexually transmitted diseases caused by bacteria only
   a) syphilis, gonorrhoea and candidiasis
   b) syphilis, chlamydiasis and gonorrhoea
   c) syphilis, gonorrhoea and trichomoniasis
   d) syphilis, trichomoniasis, pediculosis
4. Klinfelters syndrome is characterised by a karyo type of
   a) XYY b) XO c) XXX d) XXY
5. The total no. of nitrogenous base in human genome is estimated to be about
   a) 3.5 million b) 35000 c) 35 million d) 3.1 billions
6. Modern man belongs to which period
   a) Quatenary b) Cretaceous c) Silurian d) Cambrian
7. Match the pathogens with respective diseases caused by them and select the correct match using the codes given below

(A) leishmania donavani	-	(i) Amoebiasis
(B) Wucheriria bancrafti	-	(ii) Kala - azar
(C) Trypanosoma gambiensee	-	(iii) Sleeping - Sickness
(D) Entamoeba histolytica	-	(iv) Filariasis

   a) A-i B-iv C-iii D-ii b) A-ii B-iv C-i D-iii
   c) A-iii B-i C-ii D-iv d) A-ii B-iv C-iii D-i
8. World bio fuel day is
   a) August 10 b) September 10 c) July 10 d) January 10

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Section - II

II **Answer any four of the following questions.** **4x2=8**

9. What is vaccine? What are its types?
10. What is paedogamy? Give Example.
11. Write any two applications of Karyotyping.
12. Differentiate - Template strand and coding strand.
13. Draw and label the structure of the human ovum.
14. List out the major gases seems to be found in the primitive earth.

Section - III

III **Answer any three of the following questions. (Q.No.19 is compulsory)** **3x3=9**

15. Expand the acronyms.
    a) FSH b) LH c) hPL
16. Explain the mode of sex determination in honey bees.
17. State any three goals of human genome project.
18. Differentiate between divergent evolution and convergent evolution with one examples for each.
19. Differentiate prebiotics and probiotics.

Section - IV

IV **Answer all the questions.** **2x5=10**

20. a) What is referred to as industrial alcohol Briefly describe its preparation.

(OR)

b) Explain the structure of Immuno globulin with a suitable diagram.

21. a) If a Marriage between colour blind man and normal visioned woman.
    i) Write the resultant of F_1 and F_2 generation.
    ii) Draw the flow chart of the cross
    iii) Write the name of the Inhexitance
    iv) The ratio of the F_2 generation.

(OR)

b) What is meant by Assisted Reproductive Technology. Explain the following.
    i) Intra-Uterine In semination ii) Surrogacy

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